GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2024 **ORDINARY LEVEL**

Centre Number	
Centre Name	
Candidate Identification Number	
Candidate Name	

Mobile phones are NOT allowed in the examination room.

MULTIPLE CHOICE QUESTION PAPER

Duration: One and a Half Hours

INSTRUCTIONS TO CANDIDATES

Read the following instructions carefully before you start answering the questions in this paper. Make sure you have a soft HB pencil and an eraser for this examination.

1. USE A SOFT HB PENCIL THROUGHOUT THE EXAMINATION.
2. DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Before the examination begins:

3. Check that this question booklet is headed

"ORDINARY LEVEL" **– 0570 MATHEMATICS 1**

4. Fill in the information required in the spaces above.
5. Fill in the information required in the spaces provided on the answer sheet using your HB pencil:
   Candidate Name, Exam Session, Subject Code and Candidate Identification Number.
   Take care that you do not crease or fold the answer sheet or make any marks on it other than those asked for in these instructions.

How to answer the questions in this examination

6. Answer **ALL** the **50** questions in this Examination. All questions carry equal marks.
7. Non-programmable Calculators are allowed.
8. Each question has FOUR suggested answers: **A, B, C** and **D**. Decide which answer is appropriate. Find the number of the question on the Answer Sheet and draw a horizontal line across the letter to join the square brackets for the answer you have chosen.
   For example, if **C** is your correct answer, mark **C** as shown below:



9. Mark only one answer for each question. If you mark more than one answer, you will score a zero for that question. If you change your mind about an answer, erase the first mark carefully, then mark your new answer.
10. Avoid spending too much time on any one question. If you find a question difficult, move on to the next question. You can come back to this question later.
11. Do all your rough work in this booklet using the blank spaces in the question booklet.
12. **At the end of the examination, the invigilator shall collect the answer sheet first and then the question booklet. DO NOT ATTEMPT TO LEAVE THE EXAMINATION HALL WITH IT.**

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1. Given that $36 \div 9 = 4$, 4 is known as the
   A Remainder
   B Divisor
   C Quotient
   D Dividend

2. The next prime number greater than 23 is
   A 31
   B 25
   C 27
   D 29

3. The absolute value of $-1-9$ is
   A -10
   B 10
   C -8
   D 8

4. Converting 0.15 to a fraction in its simplest form gives
   A $\frac{3}{20}$
   B $\frac{15}{100}$
   C $\frac{3}{100}$
   D $\frac{15}{20}$

5. Expressing 0.0597 in standard form gives
   A 5.97×10^{2}
   B 5.97×10^{-2}
   C 6.0×10^{-2}
   D 6.0×10^{2}

6. The number of significant figures in 14.501 is
   A 5
   B 2
   C 4
   D 3

7. Mary went to bed at 11: 20 pm and asked John to wake her up after 2 hours. The time by the clock that John woke up Mary is
   A 13:20 pm
   B 1:20 pm
   C 1:20
   D 13:20

8. An article is sold at a loss of 10%. Given that the selling price is 7,200 FCFA. The cost price in FCFA is
   A 7,920
   B 6,480
   C 8,800
   D 8,000

9. Given that six textbooks each of the same price cost 30,000FCFA. The cost in FCFA of three textbooks is
   A 5,000
   B 15,000
   C 90,000
   D 25,000

10. The scale of a map is 1:50,000. The actual length in km of a road measuring 4cm on the map is
   A 2
   B 20
   C 2.5
   D 25

11. Given the sets A and B, such that $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$. The sets A and B are called
   A Equal sets
   B Power sets
   C Equivalent sets
   D Subsets

12. Given that P and Q are disjoint sets. The relationship between P and Q in set notation is
   A $P \cup Q = \phi$
   B $n(P \cap Q) = 0$
   C $P \cap Q = (\phi)$
   D $P = Q$

13. The shaded region in figure 1 is represented as

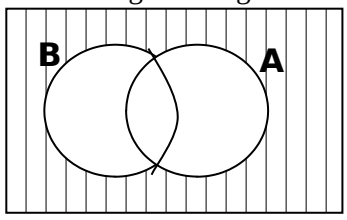


Figure 1

   A $A \cup B$
   B $(A \cap B)'$
   C $(A \cup B)'$
   D $A' \cup B'$

14. The negation of " Some students will pass the examination" is
   A No student will pass the examination
   B No student will not pass the examination
   C All students will not pass the examination
   D All students will pass the examination

15. Figure 2 is an arrow diagram of a relation between P and Q. The range of the relation is

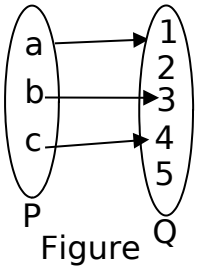


Figure 2

   A $\{2,5\}$
   B $\{1,2,3,4,5\}$
   C $\{a,b,c\}$
   D $\{1,3,4\}$

16. Given the function, $f(x) = 1 - 2x$. The value of $f(-3)$ is
   A -7
   B 7
   C -5
   D 5

17. A and B are two sets such that $A = \{3,4\}$ and $B = \{x,y\}$, the cartesian product $A \times B$ is
   A $\{(3,x),(4,y)\}$
   B $(3x,4y)$
   C $\{(3,x),(3,y),(4,x),(4,y)\}$
   D $(3,x),(3,y),(4,x),(4,y)$

18. The function f is defined on $\Re$, the set of real numbers by $f: x \mapsto 3x - 2$, $f^{-1}: x \mapsto$
   A $\frac{x+2}{3}$
   B $\frac{x-2}{3}$
   C $\frac{x+3}{2}$
   D $\frac{x-3}{2}$

19. A polygon having nine sides is called
   A Octagon
   B Decagon
   C Nonagon
   D Heptagon

20. The sum of the three interior angles of a triangle is
   A 90^0
   B 270^0
   C 360^0
   D 180^0

21. The number of lines of symmetry in a rhombus is/are
   A 8
   B 4
   C 1
   D 2

22. Figure 3 is a circle with center O and a chord PQ subtending an angle of 70^0 at the centre, O. The value of the angle marked θ is

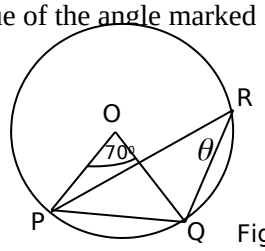


Figure 3

   A 35^0
   B 70^0
   C 55^0
   D 20^0

23. Figure 4 represents the net of


Figure 4

   A a rectangle
   B a cylinder
   C a square
   D an open Cylinder

24. Figure 5 is a rectangular based Pyramid. The number of edges is:

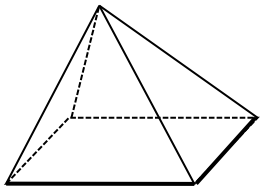


Figure 5

A 5
B 4
C 8
D 9

25. Figure 6 is a circle with centre O. The shaded region in the figure is called a

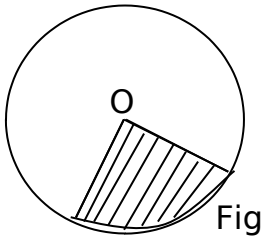


Figure 6

A Segment
B Sector
C Chord
D Secant

26. The perimeter of a rectangle with length x units and width y units in terms of x and y is

A $2xy$
B $x+y$
C xy
D $2(x+y)$

27. Figure 7 is a right–angled triangle with sides 6cm, x cm and area 24 cm^2. The value of x in the figure is

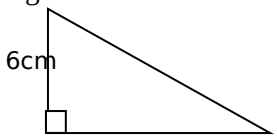


Figure 7

A 8
B 4
C 2
D 18

28. The volume in cm^3 of a cone with height 10cm and base radius 6 cm in terms of π is

A 60π
B 360π
C 120π
D 180π

29. The y-intercept of the line $y=2x+4$ on the cartesian plane is

A -4
B 2
C -2
D 4

30. The gradient of the line $3y=6x-3$ is

A -1
B 6
C -2
D 2

31 The equation of the line l in the cartesian plane shown in figure 8 is

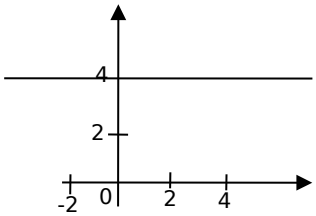


Figure 8

A $y=4$
B $x=4$
C $y=4x$
D $4y=x$

32. Subtracting $2+b$ from $2a+b$ gives

A $2-2a$
B $2a+2$
C $2a-2$
D $2a+2b-2$

33. The value of the expression x^2-y when $x=3$ and $y=2$ is

A 1
B 7
C 11
D 4

34. The range of values of x for which $7-2x\leq 19$ is

A $x\geq -6$
B $x\leq -6$
C $x\leq 6$
D $x\geq 6$

35. Factorizing $x^2 + 4x + 3$ completely gives

A $(x+4)(x+3)$

B $(x+1)(x-3)$

C $(x-1)(x+4)$

D $(x+1)(x+3)$

---

36. Simplifying 2^{-3} gives

A $-\frac{1}{8}$

B $\frac{1}{8}$

C $\frac{1}{6}$

D $-\frac{1}{6}$

---

37. The next term of the sequence – 3, 1, 5, . . . is

A 11

B 14

C 9

D 10

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38. The number of even nodes in the network in figure 9 is

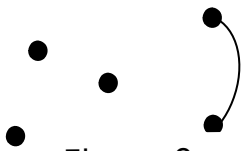


Figure 9

A 3

B 2

C 5

D 7

---

39. The relationship that exists between $\sin\theta$, $\cos\theta$ and $\tan\theta$ is

A $\tan\theta = \frac{\cos\theta}{\sin\theta}$

B $\tan\theta = \frac{\sin\theta}{\cos\theta}$

C $\tan\theta = \sin\theta + \cos\theta$

D $\tan\theta = \sin\theta - \cos\theta$

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40. Figure 10 is a right–angled triangle with sides n, 8cm and 6cm. The value of n is

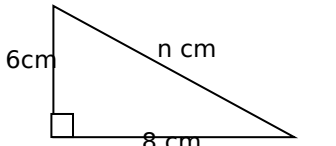


Figure 10

A 14

B 6

C 8

D 10

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41. Figure 11 shows a bearing of S from R. The bearing of R from S is

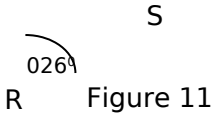


Figure 11

A 026^0

B 154^0

C 206^0

D 116^0

---

42 Given the vectors $\bar{r} = \begin{pmatrix} -3 \\ 6 \end{pmatrix}$ and $\bar{p} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, evaluating $\bar{r} + \bar{p}$ gives

A $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$

B $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$

C $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$

D $\begin{pmatrix} -4 \\ 5 \end{pmatrix}$

---

43. Given the vector $\overline{OP} = -4\mathbf{i} + 3\mathbf{j}$, $|OP| =$

A 5

B 7

C 1

D 25

---

44. The direction of the vector -3**i** -3**j** is
A 315^0
B 135^0
C 45^0
D 225^0

---

45. The order of the matrix $\begin{pmatrix} -1 & 0 & 3 \end{pmatrix}$ is
A 3×1
B 1×3
C 2×1
D 2×3

---

46. The transpose of the matrix $\begin{pmatrix} -3 & 1 & 5 \\ 0 & 2 & 7 \end{pmatrix}$ is
A $\begin{pmatrix} -3 & 0 \\ 1 & 2 \\ 5 & 7 \end{pmatrix}$
B $\begin{pmatrix} 0 & -3 \\ 2 & 1 \\ 7 & 5 \end{pmatrix}$
C $\begin{pmatrix} 5 & 7 \\ 1 & 2 \\ -3 & 0 \end{pmatrix}$
D $\begin{pmatrix} -3 & 7 \\ 1 & 2 \\ 5 & 0 \end{pmatrix}$

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47. The range of the distribution 2, 4, 2, 6, 3, 7, -1 is
A 2
B 7
C 8
D 23

---

48. The median of the distribution 7, 3, 1, 1, 4, 7,0 is
A 7
B 1
C 0
D 3

---

49. Given the distribution 8, 9, 6, 8, 15, 8, 11, 7. The mode is
A 9
B 8
C 6
D 15

---

50. The probability that Susan goes to the farm is $\frac{2}{5}$. The probability that she does not go to the farm is
A $\frac{3}{5}$
B $\frac{6}{25}$
C $\frac{2}{5}$
D $\frac{1}{5}$

STOP
GO BACK AND CHECK YOUR WORK